

Phase6 Daily Schedule

Phase 6 — Day-by-Day Schedule · Generative & Agentic AI (scientific / chemistry agents)

Weeks 58-63 · Mon Oct 11 → Sun Nov 21, 2027 · ~155 hrs

Goal: own modern LLM adaptation, multimodal, and agentic systems — reframed toward science. Wrap a model as an MCP tool, build a multi-step research/critique agent, and evaluate agents rigorously (reinforces your critic-agent doc-review project).

Blocks: **A** 06-08 · **B** 08:30-10:30 (build + threads) · **Evening** 20:30-22:30 (read, Anki, R). Weekend = buffer + rest. Threads: **T** DSA · **M** implement · **R** research/apply.

Week 58 · Oct 11-17 — Advanced LLMs / adaptation

Day	Block A	Block B (T/M)	Evening (+ R)
Mon	LoRA (low-rank adaptation)	M: implement a LoRA layer	Anki
Tue	QLoRA; quantized fine-tuning	T: DP → a QLoRA fine-tune run	Anki
Wed	PEFT trade-offs	M: compare full vs LoRA fine-tune	Anki
Thu	When to fine-tune vs prompt vs RAG	T: graphs → decision matrix	R: reproduce-paper
Fri	Adapter/prefix methods	T: timed set	Whiteboard-Fri: LoRA mechanics + when to use
Sat-Sun	Buffer + rest		

Week 59 · Oct 18-24 — Alignment (RLHF / DPO) + fine-tuning

Day	Block A	Block B	Evening
Mon	RLHF pipeline (reward model, PPO)	M: a toy reward model	Anki
Tue	DPO (direct preference optimization)	T: DP → a DPO objective	Anki
Wed	RLHF vs DPO trade-offs	M: a small preference fine-tune	Anki
Thu	Instruction tuning; data quality	T: graphs → dataset prep	R: write-up
Fri	Alignment failure modes	T: timed set	Whiteboard-Fri: RLHF vs DPO
Sat-Sun	Buffer + rest		

Week 60 · Oct 25-31 — Multimodal / VLM

Day	Block A	Block B	Evening
Mon	CLIP (contrastive image-text)	M: a small CLIP retrieval	Anki
Tue	LLaVA (visual instruction)	T: DP → run a VLM on samples	Anki
Wed	Flamingo (cross-attention fusion)	M: image+text embedding align	Anki
Thu	Multimodal for imaging+text / graph+text	T: graphs → a molecular-graph+text demo	R: reproduce-paper
Fri	VLM failure modes in science	T: timed set	Whiteboard-Fri: how CLIP aligns modalities
Sat-Sun	Buffer + rest		

Week 61 · Nov 1-7 — Agents + MCP

Day	Block A	Block B	Evening
Mon	Agent patterns (Anthropic)	M: a routing agent	Anki
Tue	Tool use; function calling	T: DP → tool schema + calls	Anki
Wed	MCP (expose models/tools)	M: wrap your property/UQ model as an MCP tool	Anki
Thu	Orchestrator-worker; evaluator-optimizer (critic loop)	T: graphs → a critic/verifier loop	R: write-up
Fri	Memory & state (LangGraph)	T: timed set	Whiteboard-Fri: MCP tool-exposure end-to-end

Sat-Sun	Buffer + rest		
---------	---------------	--	--

Week 62 · Nov 8-14 — Agent/critique evaluation + 🎯 Capstone 4 build

Day	Block A	Block B	Evening
Mon	Evaluating agents (task success)	M: build the assistant skeleton (RAG+tools)	Anki
Tue	Evaluating a critic (faithfulness/self-consistency)	T: DP → add the critic loop	Anki
Wed	Grounding & citations over literature	M: wire PMC-OA / arXiv retrieval	Anki
Thu	Failure analysis	T: graphs → eval harness for the agent	R: reproduce-paper
Fri	When an agent beats a single call	T: timed set	Whiteboard-Fri: how you'd evaluate the critic
Sat-Sun	Buffer + rest		

Week 63 · Nov 15-21 — 🎯 Capstone 4 finish + self-test

Day	Block A	Block B	Evening (+ R)
Mon	Polish the agent design	Finish the MCP scientific assistant (public data only)	Anki
Tue	Add molecular-property + UQ tool (ties 5E/5F)	Integrate + test	Anki
Wed	Eval + write-up	Run evals; document	R: write-up
Thu	Repo hardening	README + demo	R: post
Fri	🔧 PHASE-6 SELF-TEST (design an agent with a critic loop + how to evaluate the critic; MCP end-to-end; RLHF vs DPO; when an agent beats a single call)	Final polish	▶ Tier-2 applications continue
Sat-Sun	Buffer + rest · <i>Deliverable:</i> Capstone 4 (MCP scientific assistant, public)		

End of Phase 6 → Phase 7 (Computer Vision expert) next.